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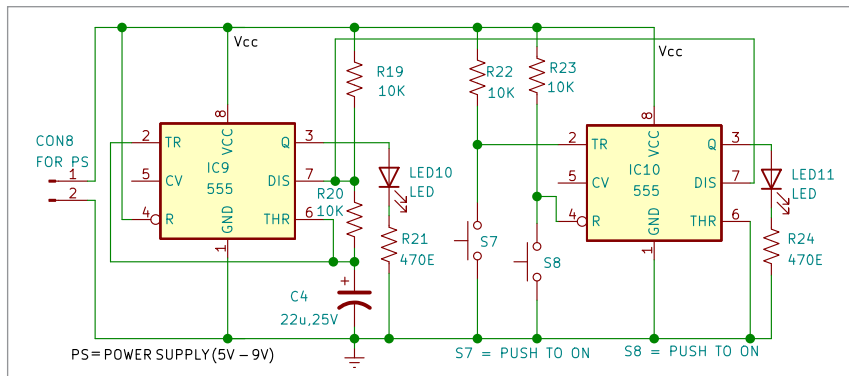


Fig. 6: Circuit diagram for experiment 6

switch S5 is pushed, LED9 turns on and stays on continuously. At the same time, the monostable operation of IC7 is resumed and LED8, which was continuously on initially, turns off after the predetermined time decided by resistor R14 and capacitor C3.

When switch S6 is pushed, LED9 turns off. This is a normal bistable operation. Thus, both LED8 and LED9 stay off in the same manner when the circuit is initially powered.

Experiment 6

Circuit diagram for experiment 6 is shown in Fig. 6. Here, two ICs, IC9 and IC10, are used. IC9 is wired as an astable multivibrator and IC10 is wired as a bistable multivibrator. In this circuit also discharge pin 7 of both the ICs are connected together.

Initially, when the circuit is powered, LED10 turns on and stays on continuously, though the circuit is wired as an astable multivibrator. This is because capacitor C4 is shorted to ground by the internal discharge transistor pin 7 of IC10. Due to this capacitor C4 cannot charge and the astable operation fails to occur.

When switch S7 is pushed, LED11 turns on and remains on until switch S8 is pushed. This is a normal bistable operation. However, when S7 is pushed, LED10 that was on continuously starts blinking on and off as the astable operation starts.

PARTS LIST

Semiconductors:

- IC1-IC10 - IC555 timer
- LED1-LED3, LED7-LED11 - 5mm white transparent LED
- LED4 - 5mm blue LED
- LED5-LED6 - 5mm bright green transparent LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R5, R12, R15, R18, R21, R24 - 470-ohm
- R2, R4, R6, R10, R11, R13, R14, R16, R17, R19, R20, R22, R23 - 10-kilo-ohm
- R7 - 1-kilo-ohm
- R8-R9 - 3.3-kilo-ohm

Capacitors:

- C1 - 470 μ F, 25V electrolytic
- C2, C3 - 220 μ F, 25V electrolytic
- C4 - 22 μ F, 25V electrolytic

Miscellaneous:

- CON1-CON8 - 2-pin connector
- S1-S8 - Push-to-on switch
- PS - 9V battery/0-12V variable power supply
- Bread board 1 or 2

When switch S8 is pushed, LED11 turns off and LED10, which was blinking on and off, stays on continuously again as explained earlier, until S7 is pushed again.

All these circuits can be wired on a breadboard so that the components can be reused, if required. A variable 0-12V, 1A power supply can be used for the experiments. Alternatively, you can use a 9V battery. Connectors CON1 through CON8 are used to connect power supply/battery. **EFY**



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